

Part C of the hiring assignment: Written Rationale
PeerBridge: Front End Developer & Designer Assignment

Q1. How would you approach building the context evaluator (FR-01 to FR-05)? What signals would you use on each target page, and what are the main risks of getting the relevance scoring wrong?

A: The scorer runs in the background service worker and only receives anonymised signals, content scripts strip PII before anything leaves the DOM.

Per page type:-

Search: Query string only. I score on length, goal-intent keywords ("should I", "career change", "how do I decide"), question structure, and word count. "Best pizza" scores near zero. "Should I quit my job to start a business at 35" scores above 0.80.

AI chat: Message count is the primary signal, FR-03 sets the floor at 3. I layer in session duration and frustration keywords ("that's not what I asked", "you keep saying the same thing"). Frustration signals carry double weight, the clearest indicator the AI has peaked.

Forms: Time-on-page and field count only. I read DOM structure, never field values. URL and page title add context, "immigration" and "mortgage" pages score higher than generic signups.

Main risks:-

False positives are existential. A card appearing during casual browsing feels like an ad, users uninstall immediately. 0.60 is a conservative floor deliberately. The less obvious risk: DOM selector fragility on AI chat pages. ChatGPT and Claude update their interfaces regularly. Selector-based message counting breaks silently on the next deployment. This needs integration tests against recorded DOM snapshots, run on every release.

Q2. What are the top two UX risks in this product, and how would you address them in your design?

A: Risk 1: Wrong timing destroys the first impression.

A suggestion mid-thought is an interruption, not help. Beyond the relevance score, I'd add a focus check, if the user typed in the last 8 seconds, hold the trigger. Wait for a natural pause. The spec doesn't require it but it's the difference between "helpful" and "annoying."

Risk 2: The "connect" handoff is the product's most fragile moment.

The user has decided to reach out to a real person, that's a vulnerable decision. If the peer is unavailable, if the context token is stale, if the destination page loses their session, the product fails at its most important interaction. FR-25 mentions the fallback in one line. It deserves the same design attention as the success path.

Q3. Which requirement would you push back on, and why?

A: FR-19: Web Crypto encryption of goal context in chrome.storage.local.

I implemented it, AES-GCM, 256-bit key, but I'd challenge it in a product review. Chrome extension storage is already isolated by origin. The real threat model here is physical device access, but the encryption key lives in the same storage location as the encrypted data. That's not defense in depth, it's the appearance of it. The actual privacy protection is FR-05: no PII leaves the browser. That's what protects users. FR-19 adds implementation complexity and a false sense of security. If the business reason is "we want to say data is encrypted in our privacy policy," that's valid, but frame it honestly, not as a security control it isn't.

Q4. How would you test the overlay card for WCAG 2.1 AA compliance? Name at least one specific tool or method.

A: Three steps, in order:

axe DevTools: Run against the rendered card. Catches ~57% of WCAG issues automatically in under a minute.

Chrome DevTools contrast checker: Inspect each text element, check the ratio in the colour picker. I verify three pairs specifically: suggestion text on card background, button text on button background, and the muted peer summary text; the last one fails most often. Both light and dark variants need a separate pass.

Manual keyboard walkthrough: Tab into the card, cycle through all three buttons (Connect → Not now → Pause today), verify focus ring is visible at every step, press Escape and confirm dismiss. Two minutes. Catches what automated tools miss.

The card uses `role="dialog"`, `aria-label`, `aria-describedby`, and `aria-live` on the countdown, sufficient for a screen reader to announce the card meaningfully on appearance.